

Name: _____

Math 166

Day 1 Questionnaire

As a team, find the answers to the questions below in the syllabus.

0. What is your section number?
1. What is your TA's name?
2. What is your professor's name?
3. What are the names of our Learning Assistants?
4. What percentage of your grade are the WebAssign online homework assignments?
5. Where do you find information about office hours?
6. What do the assignments look like for a typical week?
7. How are exams structured in this class?
8. What should you do if you are absent?

Directions: You must correctly answer 8 or more questions to pass this exam.

1. Simplify into a single fraction:

$$\frac{x}{(x+2)(x-3)} + \frac{1}{x+2}$$

2. Simplify into an expression with a single x :

$$\frac{x^2}{x^{-5/2}}$$

3. Find integers a and b where:

$$\frac{1}{x^2 - 2x + 6} = \frac{1}{(x+a)^2 + b}$$

4. Find all values of x where:

$$1 = \frac{1}{x} + \frac{2}{x^2}$$

5. For which value of a is the following true?

$$2\ln(x) + \ln(x^a) = \ln(x^5)$$

For the remaining questions, **Circle every answer** that is equivalent to the original expression, **and cross out every answer** that is not equivalent.

6. Original: $\ln(1/x)$. Options:

$$(a) : -\ln(x) \quad (b) : \ln(x^{-1}) \quad (c) : 1/\ln(x) \quad (d) : \ln(-x)$$

7. Original: $\ln(xy)$. Options:

$$(a) : \ln(x) + \ln(y) \quad (b) : \ln(x) \ln(y) \quad (c) : -\ln(1/(xy)) \quad (d) : x \ln(y)$$

8. Original: $\frac{x}{x^2+1}$ for $x \neq 0$ and $x \neq 1$. Options:

$$(a) : \frac{1}{x+1/x} \quad (b) : \frac{1}{x} + x \quad (c) : 1 - \frac{x^2 - x + 1}{x^2 + 1} \quad (d) : \left(\frac{x^2 + 1}{x}\right)^{-1}$$

9. Original: $\frac{1}{3+\sqrt{3}}$. Options:

$$(a) : \frac{1}{3} + \frac{1}{\sqrt{3}} \quad (b) : 3 - \sqrt{3} \quad (c) : \frac{3 - \sqrt{3}}{6} \quad (d) : \frac{1}{\sqrt{12}}$$

10. Original: $\sqrt{25-9}$. Options:

$$(a) : 2 \quad (b) : \sqrt{25} - \sqrt{9} \quad (c) : 4 \quad (d) : \sqrt{16}$$

Directions: Compute the following derivatives. You must correctly answer 8 or more questions to pass this exam.

1. $\frac{d}{dx} x^8 + 3x^2 + e$ _____

2. $\frac{d}{dx} e^{2x} \sin x$ _____

3. $\frac{d}{dx} \frac{2}{x^{1/2}}$ _____

4. $\frac{d^2}{dt^2} \cos(2t)$ _____

5. $\frac{d}{dx} \ln(x^{-2})$ _____

6. $\frac{d}{du} u \sec u$ _____

7. $\frac{d}{dx} \tan(x)$ _____

8. $\frac{d}{dx} (4x - 21)^{13}$ _____

9. $f'(x)$ if $f(x) = \sin(\cos(x))$ _____

10. $\frac{d}{dx} \left(f(x) + g(x) \right)$ in terms of $f(x)$ and $g(x)$ _____